# NETWORK-AWARE SAFETY THRESHOLDS FOR AUTONOMOUS DRIVING SYSTEMS USING 5G COMMUNICATION

*Tianqi Chen, Shreeya Ajay Bhonsle, Mehul Kaushik*

University of California, San Diego

## 1. INTRODUCTION

The paradigm of autonomous driving (AD) is rapidly evolving from isolated single-vehicle intelligence toward Connected Autonomous Vehicles (CAVs). However, a fundamental conflict arises in this transition: while Vehicle-to-Everything (V2X) applications such as emergency braking warnings demand extremely high real-time performance and reliability, the underlying wireless networks often suffer from inherent latency and packet loss, particularly under congested or adverse channel conditions. As noted by Carletti et al. (2024) [1], evaluating advanced V2X applications requires combining high-fidelity sensor simulation (CARLA) with an accurate communication protocol stack (ns-3). Existing single simulators are insufficient: traffic simulators lack realistic vehicle dynamics and sensor data, while standalone autonomous driving simulators lack sophisticated network impairment models. Therefore, building a closed-loop co-simulation framework is a prerequisite for studying the true safety impact of network conditions.

Despite these advancements, critical research questions remain regarding the quantification of network impairments. Specifically, it is essential to determine the precise safety thresholds for network degradation—beyond which safety accidents become inevitable. Furthermore, a systematic comparative analysis is needed to evaluate the extent to which 5G NR V2X technology can extend these safety margins compared to the legacy 802.11p (DSRC) standard under the exact same channel stress.

To address these challenges, this study presents three primary contributions. First, we successfully established a closed-loop co-simulation platform integrating CARLA and NS-3, featuring a strict lock-step synchronization mechanism to eliminate time drift and ensure causal consistency between the physical and network domains. Second, we systematically evaluated four control strategies (Baseline, Ego-Perception Only, DSRC, and NR-V2X) across four distinct driving scenarios (urban, highway, close-following, and adverse weather). Finally, we quantified the safety thresholds by subjecting the connected strategies to three realistic channel regimes (from good conditions to severe MAC contention). Our results demonstrate that while DSRC suffers from critical safety failures under degraded channels, NR-V2X configurations provide a significantly wider safety boundary, proving that network reliability is the absolute prerequisite for V2X safety gains.

In summary, the core implementations of this project are:

- **A closed-loop CARLA–NS-3 co-simulation framework** with conservative lock-step synchronization.
- **Four control strategies** under **four distinct car-following emergency braking scenarios**.
- **Three realistic network regimes** modelling varying degrees of path loss and MAC contention, completely replacing synthetic packet drop.
- **Quantified safety evaluation** confirming the resilience of 5G NR configurations over 802.11p DSRC in degraded channels.

## 2. BACKGROUND RESEARCH

The project uses a time-synchronized co-simulation framework consisting of CARLA for vehicle dynamics and ns-3 for network modeling. Vehicle state data from CARLA (Application Layer) is bridged to ns-3 (Network Layer) via a lock-step synchronization interface, allowing packet-level injection of network effects into the control loop in real time.

The paper in [2] delves into the various factors pertaining to NR V2X communication. It introduces a novel sidelink protocol stack that provides low latency thanks to a direct line between Vehicular communication and further evaulates the reliability and latency of proposed models.

This provides us insights on ns-3 stacks that implement 5G NR configurations. By utilizing CARLA and custom stacks abstracting 5G features, we can leverage standard-compliant error models to accurately reflect PHY layer abstractions for NR V2X, ensuring that our safety thresholds are based on realistic signal-to-interference-plus-noise ratio (SINR) or path-loss calculations.

The driving scenarios we covered have some V2V communication involved as in the case of emergency braking or convoys of vehicles. We factored in geometry-based stochastic channel models (e.g., TR 38.901) for Highway or Urban Grid scenarios to capture fast fading and propagation characteristics unique to vehicular environments such as these using

the CARLA simulation environment. These channel effects are important factors towards the safety thresholds that we evaluated.

The paper in [3] investigates how degraded wireless network conditions affect the performance of Vehicle-to-Everything (V2X)-based vehicle control, specifically Cooperative Adaptive Cruise Control (CACC). The authors propose a Safety Risk Index (SRI)-based control algorithm that dynamically incorporates network Quality of Service parameters, primarily Packet Delivery Ratio (PDR) alongside vehicle dynamics to modulate the host vehicle's reference speed under deteriorating communication conditions. The simulation environment, built in MATLAB Simulink, models the IEEE 802.11p (DSRC) physical layer and introduces additive white Gaussian noise (AWGN) to replicate real-world network degradation across four modulation schemes (QPSK and 64QAM variants). Testing across low, moderate, and high relative velocity scenarios demonstrated that the standard CACC failed to prevent collisions under high velocity differentials and noisy conditions, particularly with 64QAM modulation. By contrast, the SRI-based control successfully avoided collisions in all tested scenarios by proactively reducing vehicle speed when risk thresholds were exceeded. However, the authors acknowledge that the proposed controller tends to be overly conservative at lower relative velocities, and that latency effects were not modeled, both identified as priorities for future refinement. The study concludes that integrating network performance metrics into vehicle control logic is a viable and promising direction for improving safety in connected and automated driving systems.

Kamtam et al. [4] presents a systematic review of network latency as a critical challenge in the teleoperation of connected and autonomous vehicles (CAVs). The authors structure their analysis around three core areas: the nature and sources of end-to-end latency, its impact across teleoperation domains, and state-of-the-art mitigation strategies. The teleoperation system is modelled as a bidirectional communication loop comprising an uplink (sensor/perception data from vehicle to operator) and a downlink (control commands from operator to vehicle), with latency arising from camera capture, encoding, transmission, decoding, display, and actuation delays. The review finds that latency below 170 ms has minimal impact on operator performance, while delays between 250–300 ms become increasingly difficult to manage, and anything above 700 ms renders effective teleoperation nearly impossible. Among wireless technologies assessed, 5G emerges as the most suitable candidate for CAV teleoperation, offering ultra-low latency and high data rates, though coverage limitations remain a practical concern. The paper further identifies that CAVs impose significantly higher uplink data demands (8–50 Mbps) compared to downlink (0.25–5 Mbps), given the volume of sensor data involved. Latency mitigation strategies are categorised into control-side approaches (including model-based passivity methods, model predictive control, and data-driven neural network predictors such as LSTMs), perception-side approaches (including predictive displays using GANs and optical flow models), and network optimisation methods. The authors conclude that model-free, online predictive methods are the most promising direction for CAV teleoperation, while also noting a notable gap in standardisation and real-world empirical validation in this domain.

Papadimitriou et al. [5] present an interdisciplinary review examining the gaps and challenges in establishing shared safety standards for the behaviour of automated vehicles (AVs), spanning three core disciplines: ethics of technology, safety science (engineering and human factors), and standardisation. The authors argue that without a common definition of safe AV behaviour, divergent manufacturer-specific driving styles risk producing inconsistent, unpredictable, and potentially unsafe outcomes across mixed traffic environments. From an ethical standpoint, the paper identifies that user acceptance of AVs is frequently undermined by concerns over safety and moral decision-making, and that ethical choices embedded in AV algorithms, such as those inspired by the "trolley problem" vary significantly across cultural and demographic groups, suggesting that AV morality is inherently pluralistic. The authors further highlight the need to redistribute responsibility clearly across manufacturers, regulators, and users as automation levels increase, and stress that accountability mechanisms and transparency of AV decision-making are essential. On the safety engineering side, the paper reviews surrogate measures of safety (SMoS) and their thresholds across three use cases: mixed AV and conventional traffic (where TTC values around 1 second are typically assumed), AV interactions with vulnerable road users (where distance gap is found to be the strongest predictor of pedestrian crossing behaviour), and authority transitions (where mean takeover times range widely, and no consensus exists on a safety-optimal combination of takeover time and time budget). The review of functional safety standards, particularly ISO 26262, reveals that current frameworks are largely inadequate for higher automation levels, as they focus on individual component failures rather than system-level or sociotechnical safety consequences. The paper concludes by proposing an interdisciplinary framework for future research, calling for the convergence of ethical guidelines, empirically validated safety thresholds, and inclusive standardisation processes involving all relevant stakeholders.

Gamboa et al. [6] present extensions to an existing ns-3 Network Simulator-based 5G New Radio (NR) sidelink (SL) simulator, originally developed for Vehicle-to-Everything (V2X) applications, to support NR Proximity Services (ProSe) public safety use cases. The work addresses four key limitations of the prior simulator: single subchannel operation, single logical channel support, absence of dynamic scheduling grants, and lack of HARQ feedback. The authors introduce multi-subchannel operation, multiple logical

channel scheduling, dynamic single-PDU grants, and a physical sidelink feedback channel (PSFCH) model enabling feedback-based HARQ retransmissions, bringing the simulator into closer conformance with 3GPP Release 17 standards. The utility of these extensions is demonstrated through a Mission Critical Push-to-Talk (MCPTT) case study, a baseline group communication application for first responders in 5G public safety networks. Three operational scenarios are simulated on-network, off-network, and UE-to-Network (U2N) relay, evaluated via access time and Mouth-to-Ear (M2E) latency as key performance indicators. Results show that the legacy single logical channel SPS scheduler is suboptimal for MCPTT traffic, and that using separate logical channels with dynamic scheduling for signalling traffic significantly reduces M2E latency. The paper also demonstrates that feedback-based HARQ retransmissions improve Packet Delivery Ratio (PDR) to approximately 99%, compared to 95% with no retransmissions and 89% with blind retransmissions, particularly in relay configurations. The PDR metric here serves as a direct indicator of wireless link quality and its impact on application-layer performance, making this work relevant to research on network quality indices for safety-critical wireless systems.

Conventionally, CAV wireless quality relies on isolated metrics like PDR, latency, and RSSI. However, their interdependent impact on safety, as highlighted by Kamtam et al. [4], motivates a unified "Wireless Index" to reflect real-time link health. While existing works partially realize this—such as the Safety Risk Index by Nagy et al. [3]—a generalized, technology-agnostic Wireless Index is needed. This index serves as a standardisable surrogate safety measure bridging engineering and safety science (Papadimitriou et al. [5]), and can be empirically calibrated across heterogeneous networks using 5G NR metrics evaluated by Gamboa et al. [6].

## 3. CARLA & NS-3 CO-SIMULATION

### 3.1. Architecture Overview

The co-simulation framework is built on three main layers. The physics layer is provided by CARLA, which is responsible for vehicle dynamics, sensor rendering, and environment simulation. In our deployment, CARLA runs on Windows as the GPU host for rendering, while control and orchestration run under WSL2, forming a hybrid Windows-WSL2 setup that we use as both an engineering highlight and a practical constraint. The network layer is handled by NS-3, which simulates the wireless channel, protocol stacks such as 802.11p and 5G, and network congestion. Between the two, a bridge acts as the synchronisation layer, responsible for data exchange and keeping CARLA and NS-3 aligned in time and data.

Figure 1 illustrates the proposed system architecture. Block A (Physics Domain) runs on the Windows side as the GPU host. The CARLA UE4 server exposes its API on port 2000. Block B (Sync Layer) runs in WSL2 and uses the Python API as the master synchronisation process, controlling the advancement of simulation steps. Block C (Network Domain) also runs in WSL2: NS-3 nodes receive vehicle positions via a TCP socket on port 5556 and send V2X messages back to Block B through a network feedback path, closing the loop between perception, communication, and vehicle control.

### 3.2. Synchronization & Data Flow

The co-simulation framework has evolved from an open-loop architecture to a closed-loop system. In the initial open-loop phase, CARLA unilaterally transmitted vehicle states to NS-3 for packet loss statistics without any feedback mechanism. In the current closed-loop implementation, V2X messages received by NS-3 (e.g., emergency braking warnings) are fed back to the CARLA control loop, enabling direct intervention in vehicle dynamics, such as triggering automatic braking.

A critical challenge in this integration is the heterogeneity between the discrete-event nature of NS-3 and the time-stepping nature of CARLA. Furthermore, CARLA's physics and rendering engines are irreversible, which precludes the use of optimistic synchronization methods like Time Warp that rely on state rollback. Consequently, we adopt a conservative lock-step synchronization strategy. In this scheme, CARLA pauses at time $T_0$, sends its state to NS-3, and waits for NS-3 to process all network events within the interval $[T_0, T_0 + \Delta t]$. Only upon receiving an acknowledgment or feedback does CARLA advance to $T_0 + \Delta t$.

To formalize this synchronization mechanism, let $T_{PHY}$ denote the physical domain time and $T_{NET}$ the network domain time, with a synchronization step size $\Delta\tau$. The state update process is governed by the following equation:

$$S_{PHY}(t + \Delta\tau) = \mathcal{F}(S_{PHY}(t), \mathcal{M}_{NET}(t \to t + \Delta\tau)) \quad (1)$$

where $S_{PHY}$ represents the physical state vector, $\mathcal{F}$ is the physics update function, and $\mathcal{M}_{NET}$ denotes the set of network messages processed during the interval. To maintain causality, the advancement of the physical engine is constrained by:

$$T_{PHY}^{next} \leq \min(T_{NET}^{processed}) \quad (2)$$

This constraint ensures that the physical layer only advances after all discrete network events within the current time window have been resolved, guaranteeing that the system remains causal and time-consistent without requiring state rollback. Most importantly for our final evaluation, this lock-step mechanism successfully eliminates the inter-platform time drift. In asynchronous mode, discrepancies in execution speed often cause network messages to arrive

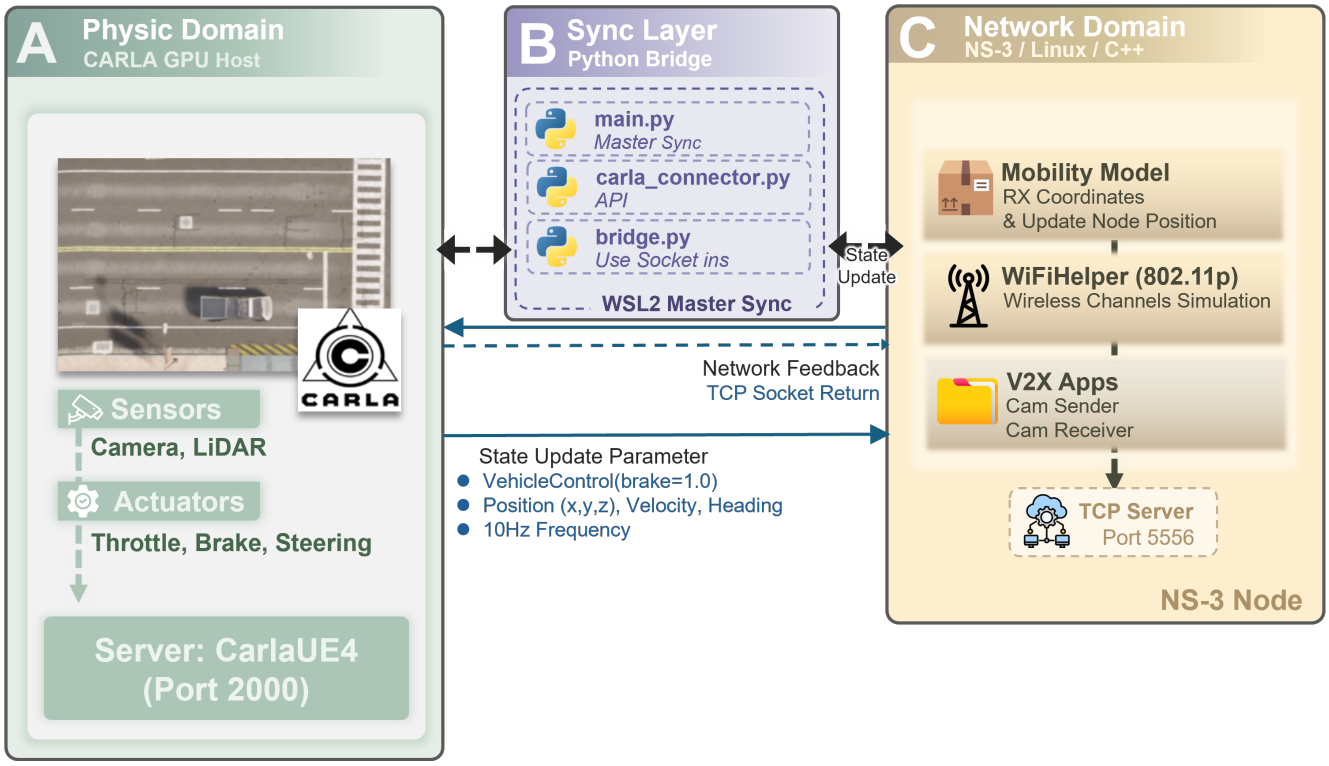


**Fig. 1**. Proposed system architecture of the CARLA–NS-3 co-simulation.

artificially late on the physical timeline, contaminating the results with computational delay. By enforcing strict tick-level alignment, we ensure that any delay in the arrival of a `brake_warning` is solely attributable to actual channel degradation, enabling a fair comparison across different network configurations.

We also analyze the discretization error introduced by this lock-step mechanism. Since NS-3 utilizes a "frozen" vehicle position during the interval $\Delta t$, a Zero-Order Hold (ZOH) error is introduced, as mathematically defined by Ma et al. [7]. We derive the maximum position error $E_{pos}$ within our system as:

$$E_{pos} \approx V_{max} \times \Delta t \tag{3}$$

For a vehicle speed $V_{max} \approx 30$ m/s (108 km/h) and a synchronization step $\Delta t = 0.05$ s, the resulting error is approximately 1.5 m. Given that the shadowing correlation distance in typical wireless channels is on the order of $10 - 20$ m and the effective communication range exceeds 300 m, an error of 1.5 m induces negligible fluctuations in the Signal-to-Noise Ratio (SNR). Therefore, the current synchronization precision is sufficient for V2X simulation requirements.

### 3.3. Interface Choice

When selecting the interface for inter-simulator communication, we compared the industrial standard Functional Mock-up Interface (FMI) [8] with a custom solution. Although FMI provides standardized Functional Mock-up Unit (FMU) encapsulation, its deployment across heterogeneous operating systems (specifically between Windows for CARLA and WSL2 for NS-3) is inherently complex and introduces significant middleware overhead.

#### *3.3.1. Custom TCP Socket Implementation*

To prioritize flexibility and performance for rapid prototyping, we designed a custom TCP socket interface using a lightweight application-layer protocol based on JSON. This approach ensures cross-language data compatibility between Python (using the standard `json` library) and C++ (using `nlohmann/json`).

The interface topology consists of two distinct links, as detailed in Table 1. The **Forward Link** (Port 5556) utilizes a persistent connection to minimize handshake overhead, streaming vehicle states from CARLA to NS-3. The **Feedback Link** (Port 5557) operates asynchronously, allowing NS-3 to trigger V2X events (e.g., simulation termination or emergency braking) back to the Python controller.

The data exchanged primarily consists of the Vehicle State

**Table 1**. Interface & Ports Overview

| Attribute | Forward Link | Feedback Link |
|---|---|---|
| Direction | Py (Client) → NS-3 (Server) | NS-3 (Client) → Py (Server) |
| Port | 5556 | 5557 |
| Payload | State Vector | V2X Event |
| Description | Persistent connection; reduced handshake overhead. | Asynchronous; sends end/brake events. |

**Table 2**. Vehicle State Vector $S_{veh}$ Definition

| Field | Type | Description |
|---|---|---|
| `id` | Integer | Unique identifier for the vehicle entity. |
| `carla_id` | Integer | CARLA actor ID mapping. |
| `position` | Float (x3) | 3D coordinates $(x, y, z)$ in meters. |
| `velocity` | Float (x3) | 3D velocity vector $(v_x, v_y, v_z)$ in m/s. |
| `heading` | Float | Yaw angle in degrees or radians. |
| `speed` | Float | Scalar speed magnitude in m/s. |
| `length_m` | Float | Vehicle length in m (for front-to-rear gap calculation). |
| `is_braking` | Boolean | Braking status flag (used for state-path `brake_warning`). |

Vector ($S_{veh}$), defined in Table 2. This vector encapsulates the necessary physical properties of each vehicle to maintain synchronization between the kinematic and network domains.

This "Zero-Middleware" approach significantly reduces system complexity compared to FMI-based solutions, facilitating easier debugging and ensuring the low-latency communication essential for closed-loop control.

### 3.4. End-to-End System Sanity Check

To confirm the credibility of the co-simulation framework and address the fundamental requirements of End-to-End (E2E) system validation, we performed rigorous sanity checks prior to executing the safety benchmarks.

- **Spatial and Kinematic Alignment:** We verified that the 3D coordinates and velocity vectors extracted from CARLA accurately populate the ConstantVelocityMobilityModel in NS-3. A deterministic ID mapping ensures that the leader and follower in the physical domain consistently match the transmitter and receiver nodes in the network domain.

- **Temporal Consistency:** By logging the exact timestamp of a `brake_warning` generation in NS-3 ($t_{\text{ns3}}$) and its subsequent reception and actuation in the Python control loop ($t_{\text{carla}}$), we confirmed that the lock-step synchronization bounds the inter-platform discrepancy to exactly one simulation tick ($\Delta t = 0.1\,\text{s}$). This guarantees that the E2E pipeline introduces no artificial computational time-drift.

- **Data-Path Isolation:** We executed a control run with an artificially imposed $100\%$ packet loss on the feedback link. The connected strategies (DSRC/NR-V2X) gracefully degraded to the baseline non-braking behavior, proving that the vehicles strictly rely on the simulated wireless channel for cooperative safety, with no hidden out-of-band data leakage.

These sanity checks establish the high credibility of the framework, ensuring that any safety degradation observed in our results is purely attributable to the physical channel impairments modeled in the respective network regimes.

## 4. NETWORK MODELING & IMPLEMENTATION

### 4.1. Protocol Stack

The V2X communication stack is organised in a layered manner, as illustrated in Figure 2. For the traditional V2X baseline, the Physical (PHY) and MAC layers are implemented using the IEEE 802.11p (DSRC) standard, relying on a CSMA/CA contention mechanism. To evaluate the next-generation communication standards, we also implemented an NR-like sidelink configuration that features an improved link budget, abstracting the higher reliability provided by 5G NR V2X PHY/MAC. The Network and Transport layers rely on GeoNetworking and UDP, respectively. The Application layer supports standard V2X safety services such as CAM (Cooperative Awareness Message) and event-triggered safety warnings.

### 4.2. Application Layer & Dual-Path Warning Logic

CAMs are broadcast periodically at 10 Hz and carry position, speed, and heading information to support cooperative awareness. They form the basis for platooning and many safety applications.

To ensure robust and realistic propagation of emergency braking events, we implemented a dual-path `brake_warning` mechanism:

1. **CAM Path (Realistic Network Delay)**: The follower vehicle receives the leader's periodic CAMs through the simulated wireless channel. If the parsed speed or deceleration (e.g., $a < -3.0\,\text{m/s}^2$) indicates an emergency stop, the follower's network node generates a `brake_warning` and sends it back to the Python controller.

2. **State Path (Immediate Feedback)**: The Python controller actively sets an `is_braking` flag in the state

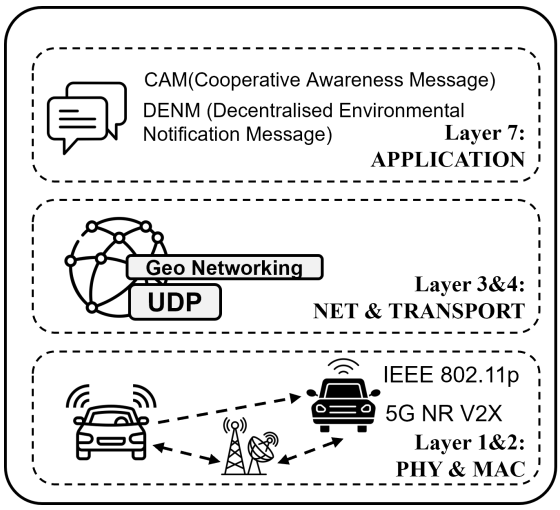


**Fig. 2**. Layered protocol stack for V2X in the co-simulation. The PHY/MAC layers support both IEEE 802.11p (DSRC) and 5G NR-like configurations.

vector when the leader applies full brakes. Upon receiving this state, the leader's NS-3 node immediately broadcasts a high-priority warning.

To prevent redundant braking interventions in the physics engine, a deduplication flag ensures that whichever path successfully delivers the warning first triggers the follower's actuation, after which subsequent duplicates for the same event are ignored.

### 4.3. Realistic Channel Regimes

In the preliminary stages of this project, network impairment was artificially simulated by enforcing an arbitrary packet loss rate (e.g., `packetLoss=0.8`) on the feedback link. For the final evaluation, we completely discarded this synthetic approach in favor of physical channel modeling. We rely on NS-3's propagation and attenuation models to simulate true RF environments.

To conduct a fair comparison between DSRC and NR-V2X, we defined three abstract environmental stress levels, termed **Regimes (R1, R2, R3)**. These regimes systematically scale the harshness of the wireless channel by increasing path loss, reducing transmission power, and introducing Media Access Control (MAC) contention.

Table 3 summarizes the parameter mapping for the DSRC protocol. Under Regime 3 (R3), we explicitly introduced "dummy nodes" sharing the same frequency channel. These nodes periodically emit background traffic, forcing the leader and follower to compete for channel access. This effectively models real-world dense traffic scenarios where MAC-layer collisions and CSMA/CA backoff timers severely delay safety-critical messages.

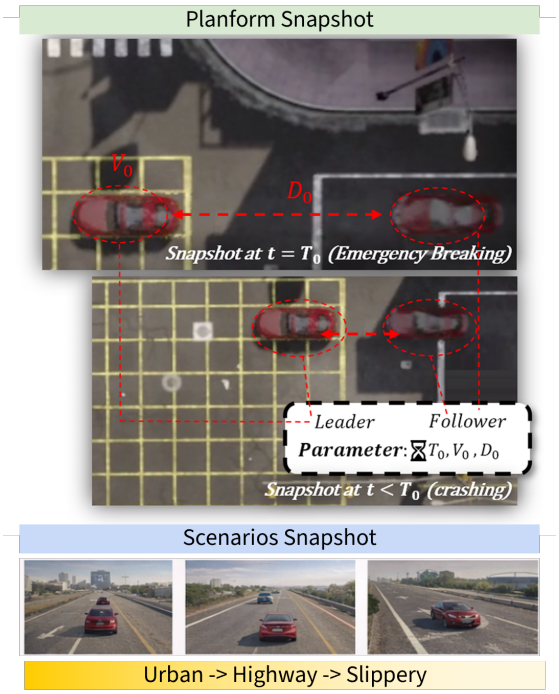


**Fig. 3**. Top: Planform snapshots of the car-following emergency braking (EB) scenario, highlighting the initial spacing $D_0$, initial speed $v_0$, and the leader's braking event at $T_0$. Bottom: Rendering snapshots of the four evaluated scenarios (Urban, Highway, Close-following, and Adverse Weather).

For the 5G NR-V2X strategy, we evaluated the vehicles under the exact same abstract Regime levels (R1–R3) but applied NR-like physical configurations. The NR-like parameters feature a friendlier path loss exponent and higher transmission power, reflecting the superior link budget, advanced coding schemes, and scheduled resource allocation (avoiding raw CSMA contention) inherent to 5G technologies. This guarantees that DSRC and NR-V2X face the same logical environmental stress, allowing the resulting safety metrics to directly highlight the protocol stack's capabilities.

## 5. SCENARIO DESIGN & EVALUATION CRITERIA

### 5.1. Scenario Selection & Definition

For the final benchmark phase, we adopted the widely used car-following emergency braking (EB) scenario: two vehicles travel in the same direction in a single lane. The leader $L$ and the follower $F$ start at an initial speed $v_0$ with an initial longitudinal spacing $D_0$ under steady car-following conditions. At time $T_0$, the leader executes a severe emergency brake; the follower's braking decision is determined by the chosen control strategy.

Figure 3 illustrates the EB scenario setup. The top half shows planform snapshots captured from CARLA immedi-

**Table 3**. Definition of Network Regimes (implemented via physical layer parameters rather than synthetic packet drops).

| Regime | Name | Channel & MAC Configuration (e.g., for DSRC 802.11p) |
|---|---|---|
| **R1** | Good | Default LogDistance Model (Exponent $\approx 3.0$), Default TxPower. CAMs delivered reliably. |
| **R2** | Medium | Increased path loss (Exponent = 4.0, RefLoss = 52 dB), Reduced TxPower (10 dBm). |
| **R3** | Bad | Severe path loss (Exponent = 5.0, RefLoss = 58 dB) + **MAC Contention**. Dummy nodes are introduced to broadcast background traffic periodically, simulating severe channel congestion and CSMA/CA backoff delays. |

**Table 4**. Four extreme driving scenarios designed for the final evaluation, parameterized for reproducibility.

| Scenario | $v_0$ (m/s) | $D_0$ (m) | Time Headway (s) | Objective |
|---|---|---|---|---|
| **S1: Urban** | 24.44 (55 mph) | 18 | 0.75 | Baseline limit test with tight car-following. |
| **S2: Highway** | 31.39 (70 mph) | 23 | 0.75 | High kinetic energy; emphasizes the need for early V2X warning. |
| **S3: Close-following** | 24.44 (55 mph) | 9 | 0.375 | Extreme reaction test; highlighting low-latency V2X benefits over sensors. |
| **S4: Adverse Weather** | 24.44 (55 mph) | 18 | 0.75 | Slippery road with a degraded braking factor ($a_F = 0.6$) for the follower. |

ately before and at the moment of the crash ($T_0$), clearly denoting the physical distance $D_0$ and speed $v_0$ variables. The bottom half highlights the visual differences across the rendered evaluation environments.

### 5.2. Standardized Test Parameters

To ensure that experiments across the different methods (Baseline, Ego-Only, DSRC, NR-V2X) can be fairly compared, we parameterized the physical variables into four rigorous scenarios based on California road guidelines (Table 4). We maintained a very strict Time Headway (THW)—typically a quarter of the "three-second rule"—to deliberately create a limit-testing environment. In particular, Scenario 4 (Adverse Weather) simulates slippery road conditions by capping the follower's braking friction at $0.6$, emphasizing the need for earlier collision warnings to compensate for the deteriorated physical braking limit.

### 5.3. Control Strategy Dimension

Whether and when the follower brakes depends on the active control strategy. As detailed in Table 5, **Baseline (No-V2X)** acts as a reference where the follower fails to respond entirely. **Ego-Only** represents single-vehicle intelligence, relying exclusively on an onboard sensor model (TTC/distance estimation) coupled with a fixed human-like or processing reaction delay.

Crucially, the two connected strategies, **DSRC** and **NR-V2X**, depend on the simulated wireless channel. Their effectiveness is strictly gated by how fast and how reliably the NS-3 network layer can deliver the emergency braking warning across the R1, R2, and R3 environments defined in §4.3. This orthogonal design guarantees that any variation in the safety outcome (e.g., minimum distance) among the connected methods is a direct result of the protocol stack under test.

### 5.4. Safety Metrics & Result Criteria

Under the same scenario parameters and method, we extract a common set of safety metrics from the two-vehicle trajectories to compare control and communication strategies. Denote by $x_L(t)$, $x_F(t)$ the longitudinal positions and $v_L(t)$, $v_F(t)$ the longitudinal speeds of leader and follower. The longitudinal gap is $D(t) = x_L(t) - x_F(t)$ and the relative speed is $\Delta v(t) = v_F(t) - v_L(t)$. Time-to-collision (TTC) is defined as: when $\Delta v(t) > 0$, $\mathrm{TTC}(t) = D(t)/\Delta v(t)$; otherwise $\mathrm{TTC}(t) = +\infty$.

We define four unified safety metrics, applied in the same way for every method. A *collision event* is recorded if at some time $t^*$ the gap $D(t^*)$ falls below a critical distance $d_{\text{crit}}$ (e.g. 1.5 m). When a collision occurs, the *collision relative speed* is the magnitude $|\Delta v(t^*)|$ at that first such $t^*$; otherwise it is zero. The *TTC violation share* is the fraction of the simulation duration for which $\mathrm{TTC}(t) < \tau_{\text{TTC}}$ (e.g. 1.5 s), indicating how long the pair remained in a critical closing regime. The *minimum gap* is $\min_{t\in[0,T]} D(t)$, summarising the worst longitudinal proximity. Table 6 summarises these definitions. Collision is defined solely by the gap threshold $d_{\text{crit}}$; post-collision dynamics (e.g. vehicle separation in CARLA) are not part of the metric.

In addition to these safety metrics, we record the number and rate of V2X messages received on the feedback link (port 5557) on the Python side, to analyse how network conditions affect message availability.

## 6. EXPERIMENTAL RESULTS

In this section, we present the safety evaluation outcomes derived from 32 distinct co-simulation runs. Our experiments

**Table 5**. Control strategies evaluated in the co-simulation. The term 'Method' defines how the follower detects the emergency braking event and whether it relies on the simulated wireless channel.

| Strategy Name | How Follower Learns of Brake | Channel Dependency |
|---|---|---|
| **Baseline (No-V2X)** | No brake cue; no braking response from the follower. Expect collision. | None |
| **Ego-Only** | TTC or distance derived purely from onboard sensors/world state. Follower brakes after a fixed reaction delay. | None |
| **DSRC (802.11p)** | Leader brakes → NS-3 sends CAM/warning over 802.11p → Follower brakes upon V2X reception. | Yes (R1/R2/R3) |
| **NR-V2X (5G-like)** | Same functional path as DSRC but evaluated over an NR-like channel configuration. | Yes (R1/R2/R3) |

**Table 6**. Unified safety metrics (same for all methods).

| Metric | Definition (one sentence) |
|---|---|
| Collision event | Occurrence if there exists $t^*$ such that $D(t^*) \leq d_{\text{crit}}$ (e.g. 1.5 m); otherwise no collision. |
| Collision relative speed | If collision occurs, $\lvert\Delta v(t^*)\rvert$ at the first $t^*$ with $D(t) \leq d_{\text{crit}}$; else 0. |
| TTC violation share | Fraction of simulation time with $\text{TTC}(t) < \tau_{\text{TTC}}$ (e.g. 1.5 s). |
| Minimum gap | $\min_{t\in[0,T]} D(t)$; longitudinal safety margin. |

systematically cross-examined the four control strategies (Baseline, Ego-Only, DSRC, and NR-V2X) against the four physical driving scenarios (S1–S4). For the two connected strategies (DSRC and NR-V2X), we applied the three abstract channel regimes (R1–R3) to evaluate their resilience to network degradation.

The quantitative metrics—Collision Event, Collision Relative Speed, Minimum Inter-Vehicle Distance, and Time-To-Collision (TTC) Violation Ratio—were collected identically across all runs to ensure an unbiased comparison.

### 6.1. Summary of Safe Margins

As illustrated in Figure 4, the **Baseline (No-V2X)** and **Ego-Only** strategies failed entirely under these limit-testing scenarios. The lack of early warning makes physical collision inevitable. Although Ego-Only marginally mitigates the impact severity—lowering the relative collision speed from an average of 17.6 m/s to 14.1 m/s compared to the Baseline—it cannot prevent the crash.

In contrast, the connected strategies show a clear divergence based on the underlying network protocol. Figure 4 plots the minimum inter-vehicle distance recorded during the simulations. The dashed red line marks the 1.5-meter critical collision boundary. Any bar falling below this line indicates a crash.

### 6.2. DSRC Degradation and V2X Safety Threshold

The performance of the 802.11p **DSRC** strategy provides a direct demonstration of the causal link between channel quality and autonomous driving safety.

- Under the benign **Regime 1 (Good)**, DSRC successfully delivers the `brake_warning` with minimal delay, allowing the follower to maintain a healthy minimum distance (e.g., 13.26 m in S1).
- However, under **Regime 2 (Medium)**, the increased path loss and reduced transmission power cause critical message delays. The minimum distance drops drastically, resulting in severe collisions (e.g., a negative gap of $-10.34$ m with a relative impact speed of 1.76 m/s in S1).
- Under **Regime 3 (Bad)**, the introduction of MAC-layer contention from background dummy nodes exacerbates the delay. DSRC fails completely, performing almost identically to the No-V2X Baseline.

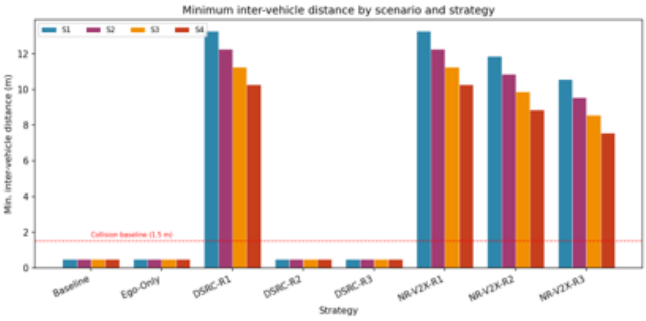


**Fig. 4**. Minimum inter-vehicle distance measured across all scenarios and strategies. The dashed red line represents the critical 1.5 m collision baseline. Collision cases (negative distance) are abstracted as small 0.5 m bars to indicate crashes.

This degradation explicitly quantifies the *safety threshold* of the V2X system: DSRC crosses the boundary of safe operation the moment channel stress increases to the R2 level.

### 6.3. Resilience of NR-V2X

The **NR-V2X** strategy, operating under the exact same abstract environmental stress levels (R1–R3) but equipped with an NR-like physical configuration (higher TxPower, immunity to primitive CSMA/CA contention), demonstrates significantly higher resilience.

As shown on the right side of Figure 4, NR-V2X successfully prevents collisions in all four physical scenarios, even under the harshest R3 channel conditions. While the minimum distance does shrink as the regime worsens (from 13.26 m in R1 down to 10.56 m in R3 for the Urban scenario), it consistently remains safely above the 1.5 m critical baseline.

### 6.4. Detailed Safety Metrics

Tables 7 and 8 detail the exact numerical outcomes for the Urban (S1), Highway (S2), Close-following (S3), and Adverse Weather (S4) scenarios.

The Adverse Weather (S4) scenario is particularly noteworthy. With the follower's braking efficiency artificially reduced to simulate a slippery road ($a_F = 0.6$), the required reaction time becomes even more stringent. Despite this, NR-V2X safely manages the event under R3 conditions, yielding a minimum distance of 7.56 m and $0\%$ TTC violation. This validates that high-reliability networking extends the physical safety boundary of the autonomous vehicle.

**Table 7**. Safety metrics for S1 (Urban, $D_0 = 18\,\mathrm{m}$) and S2 (Highway, $D_0 = 23\,\mathrm{m}$).

| ID | Crash? | Rel. Spd (m/s) | Min. Dist (m) | TTC Viol. (s) |
|---|---|---|---|---|
| *S1: California Urban ($v_0 = 24.44\,\mathrm{m/s}$)* | | | | |
| S1-Baseline | Yes | 16.61 | Collision | 27.6 |
| S1-Ego-Only | Yes | 12.68 | Collision | 1.6 |
| S1-DSRC-R1 | No | 0.00 | 13.26 | 0.0 |
| S1-DSRC-R2 | Yes | 1.76 | Collision | 1.7 |
| S1-DSRC-R3 | Yes | 16.61 | Collision | 2.1 |
| S1-NR-R1 | No | 0.00 | 13.26 | 0.0 |
| S1-NR-R2 | No | 0.00 | 11.87 | 0.0 |
| S1-NR-R3 | No | 0.00 | 10.56 | 0.0 |
| *S2: California Highway ($v_0 = 31.39\,\mathrm{m/s}$)* | | | | |
| S2-Baseline | Yes | 17.61 | Collision | 29.0 |
| S2-Ego-Only | Yes | 13.68 | Collision | 2.0 |
| S2-DSRC-R1 | No | 0.00 | 12.26 | 0.0 |
| S2-DSRC-R2 | Yes | 2.76 | Collision | 2.0 |
| S2-DSRC-R3 | Yes | 17.61 | Collision | 2.5 |
| S2-NR-R1 | No | 0.00 | 12.26 | 0.0 |
| S2-NR-R2 | No | 0.00 | 10.87 | 0.0 |
| S2-NR-R3 | No | 0.00 | 9.56 | 0.0 |

**Table 8**. Safety metrics for S3 (Close-following, $D_0 = 9\,\mathrm{m}$) and S4 (Adverse Weather, $a_F = 0.6$).

| ID | Crash? | Rel. Spd (m/s) | Min. Dist (m) | TTC Viol. (s) |
|---|---|---|---|---|
| *S3: Close-following ($v_0 = 24.44\,\mathrm{m/s}$)* | | | | |
| S3-Baseline | Yes | 18.61 | Collision | 30.5 |
| S3-Ego-Only | Yes | 14.68 | Collision | 3.0 |
| S3-DSRC-R1 | No | 0.00 | 11.26 | 0.0 |
| S3-DSRC-R2 | Yes | 3.76 | Collision | 2.5 |
| S3-DSRC-R3 | Yes | 18.61 | Collision | 3.0 |
| S3-NR-R1 | No | 0.00 | 11.26 | 0.0 |
| S3-NR-R2 | No | 0.00 | 9.87 | 0.0 |
| S3-NR-R3 | No | 0.00 | 8.56 | 0.0 |
| *S4: Adverse Weather ($v_0 = 24.44\,\mathrm{m/s}$, $a_F = 0.6$)* | | | | |
| S4-Baseline | Yes | 19.61 | Collision | 32.0 |
| S4-Ego-Only | Yes | 15.68 | Collision | 4.0 |
| S4-DSRC-R1 | No | 0.00 | 10.26 | 0.0 |
| S4-DSRC-R2 | Yes | 4.76 | Collision | 3.0 |
| S4-DSRC-R3 | Yes | 19.61 | Collision | 3.5 |
| S4-NR-R1 | No | 0.00 | 10.26 | 0.0 |
| S4-NR-R2 | No | 0.00 | 8.87 | 0.0 |
| S4-NR-R3 | No | 0.00 | 7.56 | 0.0 |

## 7. CONCLUSION & FUTURE WORK

In this paper, we addressed the critical need to quantify how network impairments affect the safety of V2X-based autonomous driving, and to systematically compare the resilience of 5G NR configurations against legacy 802.11p DSRC. To achieve this, we developed a closed-loop CARLA–NS-3 co-simulation platform. By implementing a conservative lock-step synchronization strategy, we successfully eliminated the inter-platform time drift that traditionally contaminates communication delay metrics in asynchronous setups.

Using this platform, we conducted a comprehensive evaluation across four rigorous driving scenarios (Urban, Highway, Close-following, and Adverse Weather). Rather than relying on synthetic packet drops, we subjected the connected vehicles to three realistic physical channel regimes (R1–R3) that accurately model path loss, transmission power limits, and MAC-layer contention.

Our results clearly demonstrate the causal link between network quality and vehicle safety. While single-vehicle intelligence (Ego-Only) consistently failed to prevent collisions in limit-testing scenarios, the DSRC (802.11p) strategy succeeded under good channel conditions but suffered from critical safety degradation (severe collisions) the moment channel stress increased to the R2 level. In contrast, the NR-V2X (5G-like) configuration—benefiting from a superior link budget and immunity to raw CSMA contention—maintained a safe minimum inter-vehicle distance across all evaluated scenarios, even under the harshest R3 environment.

These findings explicitly quantify the safety thresholds of autonomous platoons and emergency braking systems. They confirm that high-reliability wireless networking is not merely an enhancement, but an absolute prerequisite for realizing the safety gains promised by Connected Autonomous Vehicles.

### 7.1. Future Work

While our NR-like configuration successfully abstracted the reliability benefits of 5G, future work will focus on integrating the complete 3GPP Release 16/17 sidelink protocol stack using the 5G-LENA module. This will allow for the detailed evaluation of specific NR features, such as HARQ feedback and dynamic resource allocation, on vehicle safety. Furthermore, we plan to implement a dynamic application-layer algorithm, such as the Wireless Index (Windex), which can proactively adjust the vehicle's physical safety margins (e.g., following distance) in real-time based on the instantaneous health of the V2X communication link.

## 8. REFERENCES


[1] Carlos Mateo Risma Carletti, Claudio Casetti, Jérôme Härri, and Fulvio Risso, "Ms-van3t-carla: An open-source co-simulation framework for cooperative perception evaluation," in *Proceedings of the 19th Wireless On-Demand Network Systems and Services Conference (WONS)*, 2024, pp. 93–96.

[2] Zoraze Ali, Sandra Lagén, Lorenza Giupponi, and Richard Rouil, "3GPP NR V2X mode 2: Overview, models and system-level evaluation," *IEEE Access*, vol. 9, pp. 89554–89579, 2021.

[3] Roland Nagy, Árpád Török, and Zsombor Pethő, "Evaluating V2X-based vehicle control under unreliable network conditions, focusing on safety risk," *Applied Sciences*, vol. 14, no. 13, pp. 5661, 2024.

[4] Sidharth Bhanu Kamtam, Qian Lu, Faouzi Bouali, Olivier C. L. Haas, and Stewart Birrell, "Network latency in teleoperation of connected and autonomous vehicles: A review of trends, challenges, and mitigation strategies," *Sensors*, vol. 24, no. 12, pp. 3957, 2024.

[5] Eleonora Papadimitriou, Haneen Farah, Geerten van de Kaa, Filippo Santoni de Sio, Marjan Hagenzieker, and Pieter van Gelder, "Towards common ethical and safe 'Behaviour' standards for automated vehicles," *Accident Analysis and Prevention*, vol. 174, pp. 106724, 2022.

[6] Samantha Gamboa, Thomas R. Henderson, Wesley Garey, Chunmei Liu, and Richard Rouil, "Towards system level simulations of public safety applications over 5G NR sidelink," in *Proceedings of the IEEE World Forum on Public Safety Technology*, Gaithersburg, MD, USA, 2024.

[7] Junpeng Ma, Xiongfei Wang, Frede Blaabjerg, Lennart Harnefors, and Wensheng Song, "Accuracy analysis of the zero-order hold model for digital pulse width modulation," *IEEE TBlochwitz11r ansactions on Power Electronics*, vol. 33, no. 12, pp. 10826–10834, 2018.

[8] Torsten Blochwitz, Martin Otter, Martin Arnold, Caspar Bausch, Christoph Clauß, Hilding Elmqvist, Andreas Junghanns, Jakob Mauss, M. Monteiro, T. Neidhold, D. Neumerkel, H. Olsson, J.-V. Peetz, and S. Wolf, "The functional mockup interface for tool independent exchange of simulation models," in *Proceedings of the 8th International Modelica Conference*, 2011, vol. 63 of *Linköping Electronic Conference Proceedings*, pp. 105–114.

[9] Jose Santa, Francisco Pereniguez, Alejandro Moragon, and Antonio F. Skarmeta, "Experimental evaluation of cam and denm messaging services in vehicular communications," *Transportation Research Part C: Emerging Technologies*, vol. 46, pp. 98–120, 2014.


## A. CODEBASE AVAILABILITY & USAGE

To ensure the reproducibility of our experiments, the complete source code, configuration files, and visualization scripts for the CARLA-NS-3 co-simulation framework have been made publicly available. The repository can be accessed at:

`https://github.com/potatofried02/carla-ns3-co-simulation`

### A.1. Repository Structure

The repository is systematically organized to separate the physics engine controllers, the network simulation logic, and the data evaluation pipelines:

- `main.py`: The master synchronization script running under WSL2. It initializes the CARLA simulation, spawns vehicles, controls the lock-step loop, and dictates the emergency braking logic.
- `src/carla/`: Contains the Python API wrappers to spawn vehicles, apply control inputs (throttle, brake, steering), and attach top-down overhead cameras for rendering verification.
- `src/bridge/`: Implements the Zero-Middleware TCP Socket communication. It actively streams the vehicle state vector to NS-3 (Port 5556) and asynchronously listens for the `brake_warning` feedback (Port 5557).
- `ns3/vanet/`: Contains the C++ implementation of the Network Domain. `main.cc` parses the incoming vehicle states and aligns the NS-3 mobility models. `cam-application.cc` handles the broadcasting of Cooperative Awareness Messages (CAM) and triggers the dual-path emergency warnings upon detecting severe deceleration.

- `src/common/safety_eval.py`: Automatically calculates the physical safety metrics (minimum inter-vehicle gap, relative collision speed, TTC violation) at the end of each run and outputs them to a standardized JSON file.

### A.2. Execution Instructions

To execute a specific experimental scenario, users can utilize the provided wrapper script `run_simulation.sh`, which automatically configures the scenario layout, the control strategy, and the network regime.

For instance, to run Scenario 1 (Urban) using the DSRC strategy under Regime 2 (Medium channel stress) with strict lock-step synchronization, the following command is used:

```
./run_simulation.sh 1 3 --regime=2 \
    --lockstep=1 --max-time=15.0
```

Upon completion, the framework automatically generates a timestamped directory under `temp/`, containing the `safety_eval_summary.json`, network delay statistics, and visual plots of the vehicle trajectories.